

DSA STUDY BLUEPRINT SERIES

92. Sorted Array to Balanced BST

Converting Sorted Lists to Balanced BSTs

Section 09 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Balanced BST:** Height difference between left & right is at most 1.
- Select middle array element as root node; recurse on halves.

Memory & Execution Blueprint

Arr: [1, 2, 3] $\Rightarrow$ Root: 2 • Left child: 1 • Right child: 3
(Balanced!)

C++ Implementation Reference

Standard code layout and syntax

```
TreeNode* sortedArrayToBST(vector& nums, int s, int e) {  
    if (s > e) return nullptr;  
    int mid = s + (e - s) / 2;  
    TreeNode* root = new TreeNode(nums[mid]);  
    root->left = sortedArrayToBST(nums, s, mid - 1);  
    root->right = sortedArrayToBST(nums, mid + 1, e);  
    return root;  
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Time Complexity	$O(N)$
Space Complexity	$O(\log N)$ stack

Key Practices & Gotchas

- **Important:** This approach builds a tree with minimum height.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace BST constraint validation checks verifying node ranges `[Min, Max]`:

Active Node	Valid range limit	Comparison check	Recursive branch step
15 (Root)	$[-\infty, +\infty]$	True $(-\infty < 15 < +\infty)$	Recurse Left child with range $[-\infty, 15]$
10 (Left)	$[-\infty, 15]$	True $(-\infty < 10 < 15)$	Recurse Right child with range $[10, 15]$

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Inorder property:** BST Inorder traversals yield strictly increasing sorted lists.
- **Tree Balancing:** Balanced tree heights (AVL/Red-black) guarantee logarithmic search times.
- **Related LeetCode Problems:** #98 Validate Binary Search Tree, #700 Search in a Binary Search Tree, #108 Convert Sorted Array to Binary Search Tree.